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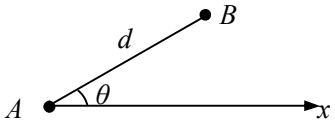
25th National High School Physics Competition: Final Problems

102008- Beijing

★ Theory

1.

In football, a player at A passes along Ax; the ball rolls on grass at speed v uniformly. A defender is at B. Let d be the A , B distance and θ the AB - Ax angle. Given $\theta < 90°$, as the ball leaves A, the defender at B starts along a straight line at speed v_p to intercept it. Ignore field boundaries. Answer the following in terms of the given parameters.



1. Find the necessary condition for interception by the defender.

2. An attacker waits on Ax, distance l_r from A. Find when the ball is not intercepted by the defender initially at B.

3. An attacker is on Ax, distance L from A. As the ball leaves A, he runs at constant speed v_r to receive it. Find when the defender initially at B does not intercept it.

2.

Ground observations give satellite motion, then the motion of spacecraft or ground objects. Both require time and spatial coordinates. To simplify analysis and calculation, ignore Earth's spin and orbit; use Earth as an inertial frame.

1. To measure satellite motion, ignore relativity. It has a radio transmitter and a precise atomic clock. Each signal contains its transmission time.

(I) Several ground stations use these data, their known positions and clocks to find its position at each instant, hence its motion. For this system,

What minimum number of ground stations is needed? Give the position equations.

(II) Stations D_1, D_2 lie on one meridian at θ N and θ S (unit: (°))
Both receive a satellite signal τ after emission. (i) Find its maximum altitude H above ground; (ii) For small latitude errors $\triangle\theta$ at D_1, D_2, find the error in H (iii) For a small error $\triangle\tau$ in time τ, find the error in H.

2. In 1 (II), take $\theta = 45°$, $\tau = 0.100$ s. (i) Find the maximum altitude H in km. (ii) If $\theta = \pm 1.0''$, how large is the error in H? (iii) If $\triangle\tau = \pm 0.010\ \mu$s, what error in H? Earth is a sphere of radius $R = 6.380 \times 10^3$ km; $c = 2.998 \times 10^8$ m/s, and ground gravitational acceleration $g = 9.81$ m/s^2.

3. Now use reference-satellite motion to track an object. Neglect relativity. Each continuously emitted radio signal states its emission time and the satellite's position then. A receiver aboard the object, with no clock, obtains these data. To infer its motion state, its position and time on receiving a signal, what minimum number of different satellites must it receive simultaneously? Write the equations for the object's position and time at reception.

4. Special relativity slows moving clocks; general relativity slows clocks in a gravitational field. Include both effects. Satellite and station clocks have aligned zero readings. Orbit height $h = 2.000 \times 10^4$ km; radius R, light speed c, and ground acceleration g are as given in part 2.

(I) By special relativity, estimate the reading difference after 24 h on Earth: ground minus satellite. Use uniform rectilinear time dilation for circular motion.

(II) The GR slowing factor is $(1-2\phi/c^2)^{1/2}$, where ϕ is the magnitude of gravitational potential: potential energy per unit mass, zero at infinity. After 24 on the ground clock, find the satellite-minus-ground clock reading.

3.

A refrigerator sends heat from a cold reservoir, plus energy from external work, to a hot reservoir, cooling the cold one and warming the hot one. If it works between absolute temperatures T_1 (hot) and T_2 (cold), taking heat

If the extracted heat is Q and external work is W, then

$$\frac{Q}{W} \leqslant \frac{T_2}{T_1 - T_2} ,$$

The sign "=" is the ideal limit. In winter, use the refrigerator as a heat pump. At outdoor temperature −5.00℃, it keeps a room at 20.00℃. The warmer room loses heat outdoors. Consider only conduction, governed by the following law. For a conducting layer of thickness l, area S, temperature difference T, the heat conducted through it per unit time from the hot side to the cold side is

$$H = k \frac{\triangle T}{l} S ,$$

Here k is the thermal conductivity, determined by the layer material.

1. Heat loss is through a pane facing out: S = 5.00 m^2 in area, l = 2.00 mm thick. The conductivity is k = 0.75 W/(m·K); electricity costs 0.50 yuan/kWh. Ideally, find the electricity cost of operating the heat pump for 12 h.

2. Replace it by double glazing: both panes 2.00 mm thick, separated by an air gap l_0 = 0.50 mm thick; conductivity k_0 = 0.025 W/(m·K). The tariff remains 0.50 yuan/kWh. For 12 h operation, how much is saved versus single glazing?

4.

Fig. 1 has metal (M), thin insulator (I) and metal (M), in close contact. Classically, if I is a perfect insulator, electrons cannot pass to another metal layer. Yet quantum physics allows this under certain conditions. Tunnelling follows from electron wave nature. Each event transfers one electron. The discrete charge is a multiple of $-e$ (e = 1.60×10^{-19} C), hence the name single-electron tunnelling. MIM is a tunnelling junction or single-electron tunnelling junction. Coulomb blockade controls it, allowing tiny single-electron devices. This active research field has promising applications.

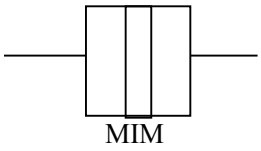


Fig. 1

1. Treat Fig. 1 as a capacitor of capacitance C, with continuous plate charge, as Fig. 2 shows: the electron cloud shifts slightly against the positive background. Tunnelling through the insulator transfers only

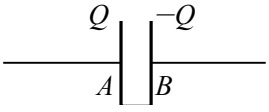


Fig. 2

discrete electron charge. Tunnelling cannot raise the electrostatic energy. This is Coulomb blockade. Find the range of plate voltage $V_{AB} = V_A - V_B$ for which single-electron tunnelling is forbidden, that is, the blockade condition.

2. If $V_{AB} = 0.10$ V just permits tunnelling, estimate the capacitance C.

3. For Fig. 1 at fixed voltage V, use the double junction in Fig. 2 to avoid stray capacitance when observing tunnelling. The middle metal layer is the single-electron island. The electrodes are S, D. Its charge is $-ne$, where n is any integer. e is unit charge; MIM capacitances are C_S and C_D.

Show that the charge-dependent (island) electrostatic energy is

$$U_n = \frac{(-ne)^2}{2(C_S + C_D)} .$$

4. In Fig. 3, add a gate to the source (S), drain (D) pair via capacitor C_G. The gate (G) forms the single-electron transistor of Fig. 4. No single electron

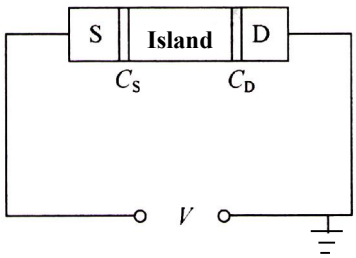


Fig. 3

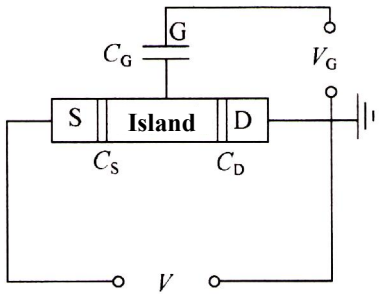


Fig. 4

tunnels between gate and island. At small fixed V, V_G controls n. How does V_G control n? Model V_G as adding the equivalent island charge $q_0 = C_G V_G$. Its energy is about $U_n = (-ne + q_0)^2 / 2C_\Sigma$, with $C_\Sigma = C_S + C_D + C_G$. On the grid (Fig. 5), allow for blockade.

Start with $n = 0$. As $C_G V_G / e$ rises from 0 to 3, draw the island's energy U_n against $C_G V_G$ in bold. Plot U_n vertically; U_n is in

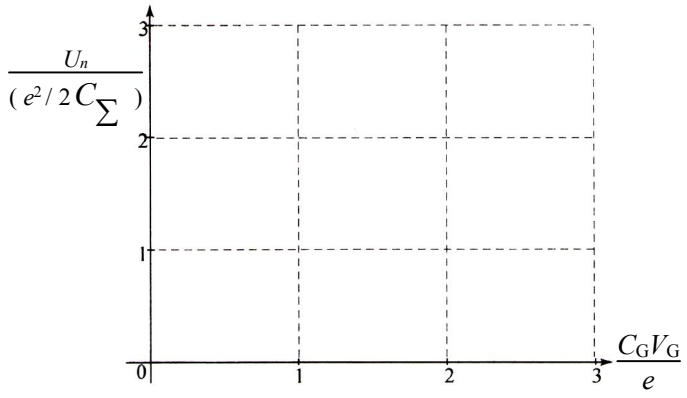


Fig. 5

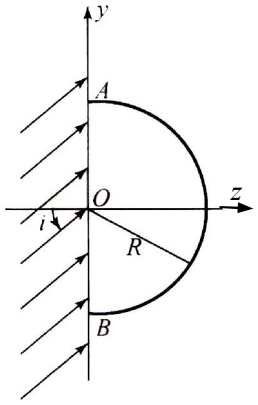
$e^2/2C_\Sigma$; use C_GV_G horizontally, with C_GV_G in units of e. Mark key points. For $n=0$, 1, 2, 3, enter C_GV_G/e ranges below. (Only the graph and Table 1 are scored.)

Table 1

n	0	1	2	3
C_GV_G/e range				

5.

A clear semicylinder, index $n=1.50$, radius R, is in air. The figure is a normal section; O is its axis intersection. Light enters only through plane AB. A broad parallel monochromatic beam, normal to the axis, covers AB at angle i. Some light passes through and exits the curved surface. Its width on AB along y is d. Exclude rays reflected one or more times inside the cylinder before exiting.



1. As incidence angle i varies over $0°\sim90°$, find the minimum d_{min} of d, and its maximum d_{max}.

2. In the figure plane, relate the arc subtended at O by the exit beam to i. Also locate this arc at grazing incidence.

6.

General relativity bends light in a stellar gravity field. The ray is in a plane through the centre, bending slightly near it. Its closest point is periapsis. A line through periapsis and centre is its symmetry axis. In polar coordinates (r, φ), take the centre as origin, symmetry axis as polar axis. The photon trajectory is

$$r=\frac{GM/c^2}{a\cos\varphi+a^2(1+\sin^2\varphi)},$$

G is the gravitational constant, M stellar mass, c light speed; a is far below 1 in magnitude. A star is 80.0 light years away; a white dwarf lies halfway to Earth. At Earth, rays past either side subtend 1.80×10^{-7} rad. Find the dwarf mass (kg).

Given $G=6.6730\times10^{-11}\ m^3/(kg\cdot s^2)$

7.

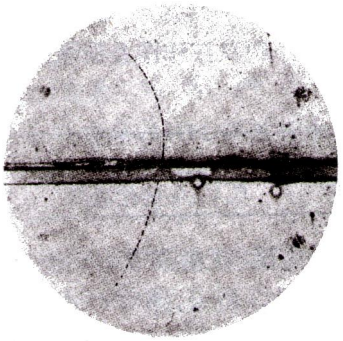
1. In the Bohr model of ground-state helium, each electron circles the nucleus, with equal radii and angular momenta $\hbar$: $\hbar = h/2\pi$, where h is the Planck constant.

(I) Ignore electron interaction. Find first ionization energy (eV): energy needed to move one electron to infinity.

(II) The measured ionization energy is 24.6eV . Include electron interaction in this model. Assume the two electrons always occupy opposite ends of a diameter through the nucleus. Find orbit radius r_0, ground energy E_0, ionization energy E^+. Compare the predicted first ionization energy with experiment.

The mass is m =0 .5MeV c^2, c is light speed; $\hbar c$ =197.3MeV7 .3eV • nm, ke^2 =1 .4MeV .4eV nm , k is Coulomb's constant and e is unit charge.

2. The photo is a cloud-chamber track, in the paper plane. In its middle a lead plate is normal to paper. A steady uniform field goes into it. The particle charge magnitude is e : e =1 .60 $\times 10^{-19}$ C. The track radius below is r_d =2 0mm , and above is r_u =7 6 .0mm . In lead, the track makes θ =1 5 .0° to the normal; the lead is d =6 .00mm thick; use B =1 .00 and m =9 .1×10 $^{-31}$ kg 0 .511 MeV / c^2. Ignore gas drag on the particle.



(I) State the direction of motion and sign of charge.

(II) Find the average force in N on the plate while the particle passes through.

(III) Instead, an accelerator pulse of flux j =5 .00×10 ^{18}/s hits the plate. One pulse lasts τ =2.5 0s . Find the average force in N on the plate while the pulse passes through, and the heat in J absorbed by the plate during it.